

This document is a short report on the queries used in the Student Management System I created incorporating the starter code

Introduction

A simple application of the SQL language to manage and analyze student data, in a database named StudentManagement. Student data – Name, Gender, Age, Grade, Subject Score is stored in the database. The data is retrieved, analysed, and updated using various SQL queries.

1. Database Creation

```
CREATE DATABASE StudentManagement;
```

Explanation:

This query will create a new database to store all student information, with a name of StudentManagement.

2. Selecting the Database

```
USE StudentManagement;
```

Explanation:

The command is used to select which database to use for the next operation and here, we are selecting the **StudentManagement** database.

3. Table Creation

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY AUTO_INCREMENT,
```

```
Name VARCHAR(50),  
Gender CHAR(1),  
Age INT,  
Grade VARCHAR(2),  
MathScore INT,  
ScienceScore INT,  
EnglishScore INT  
);
```

Explanation:

This query will form a table called Students.

Field Description:

Field Name	Description
Unique ID for each student	
Name	Student name
Gender	Gender of student
Age	Age of student
Grade	Student grade
MathScore (Maths)	Marks awarded for Mathematics
ScienceScore	Marks in Science
EnglishScore	Marks in English

Important Keywords:

PRIMARY KEY → A field that contains a value that can never be null.

AUTO_INCREMENT → sequence the StudentID for each student that is added.

4. Inserting Student Records

```
INSERT INTO Students  
(Name, Gender, Age, Grade, MathScore, ScienceScore, EnglishSc
```

```
ore)  
VALUES (...);
```

Explanation:

This is a query that adds student information to the Students table.
There are 10 student records where they get different grades and scores.

5. Show all student information.

```
SELECT * FROM Students;
```

Explanation:

Data is retrieved using the `SELECT` statement.

- means all columns.

This query returns all the information of all students that have been stored in the table.

6. Average Score in Each Subject

```
SELECT  
    AVG(MathScore) AS AvgMathScore,  
    AVG(ScienceScore) AS AvgScienceScore,  
    AVG(EnglishScore) AS AvgEnglishScore  
FROM Students;
```

Explanation:

The `AVG()` function returns the mean (average) value.

The custom column name is given by the `AS` keyword.

This query will return the mean of all students' marks in:

- Mathematics
 - Science
 - English
-

7. Top Performer Query

```
SELECT
    StudentID,
    Name,
    (MathScore + ScienceScore + EnglishScore) AS TotalScore
FROM Students
ORDER BY TotalScore DESC
LIMIT 1;
```

Explanation:

Adds all subject marks together to get the total score.

The `ORDER BY DESC` clause sorts the results from highest to lowest.

This is done using `LIMIT 1` which will display only the top student.

This query shows the student who has the highest overall marks.

8. Count the number of students per grade.Count the students in each grade.

```
SELECT
    Grade,
    COUNT(*) AS StudentCount
FROM Students
GROUP BY Grade;
```

Explanation:

- `COUNT(*)` counts records.
- `GROUP BY` groups students according to grades.

This query will display the number of students in each grade.

9. Average weighted score for each gender. Gender weighted average score.

```
SELECT
    Gender,
    AVG(MathScore) AS AvgMath,
    AVG(ScienceScore) AS AvgScience,
    AVG(EnglishScore) AS AvgEnglish
FROM Students
GROUP BY Gender;
```

Explanation:

This inquiry computes the average score for subject for these:

- Male students
- Female students

The GROUP BY Gender clause groups data by gender.

10. Students who have a Math Score above 80

```
SELECT *
FROM Students
WHERE MathScore > 80;
```

Explanation:

The WHERE clause is used to specify conditions.

This query shows those students who have marked more than 80 in Mathematics.

11. Update Student Grade

```
UPDATE Students  
SET Grade = 'A'  
WHERE StudentID = 5;
```

Explanation:

The **UPDATE** statement is used to change data in the table.

The value is changed by **SET**.

The **WHERE** clause is used to identify the record to be updated.

This is a query that will update the student's grade who has the StudentID of 5.

Conclusion

In this SQL project, you will see:

- Database creation
- Table creation
- Data insertion
- Data retrieval
- Aggregate functions
- Filtering
- Grouping
- Sorting
- Updating records

These are the basic operations of SQL and database management and data analysis.